

Ingress Gateway Evaluation

Status:  proposed architecture; no implementation decision has been applied.

Agreed Inputs and Decisions

These are evaluation inputs agreed with the platform owner, not assumptions made by the implementation.

Hard Requirements

ID	Requirement or constraint	Decision impact
R1	Open-source license suitable for commercial projects, with no mandatory paid tier or complex reciprocal obligations	Excludes proprietary-only features from the required design
R2	Mature, commonly deployed project with credible maintainers	Avoid niche GraphQL-specific gateways unless they provide a decisive benefit
R3	Private network, but production-grade security controls remain required	Internal placement does not justify relaxed TLS, identity, or header trust
R4	GraphQL queries, mutations, and graphql-transport-ws subscriptions; REST is also present	WebSocket compatibility and long-lived connections are mandatory
R5	Requests and responses may exceed 1 MiB	Body-size limits must be explicit and tested
R6	At least 100 requests/s in total; 1000 requests/s is the preferred design target	Capacity must be validated before production migration
R7	At least two replicas, with three preferred for the critical gateway data plane	The gateway is treated as critical infrastructure
R8	MetallB, TLS, Prometheus metrics, access logs, and equivalent distributed tracing	Observability cannot regress during migration
R9	Prefer standard Gateway API resources; CRDs are accepted only for missing standard features	Gateway and HTTPRoute form the portable routing layer
R10	Preserve current redirects, rewrites, CORS, authentication, and proxy behavior	Annotation inventory must be mapped route by route
R11	Keycloak and oauth2-proxy flows must retain their public issuer, callbacks, cookies, and forwarded scheme	Identity routes are migrated late and require dedicated validation
R12	No new DNS records can be created; existing names resolve through one wildcard record to the current MetallB IP	Parallel migration by DNS name or DNS weighting is unavailable
R13	No CNI replacement in this change	Cilium Gateway API is not an immediate candidate

Accepted Trade-offs

ID	Agreed position	Consequence
T1	Resource reservations may increase beyond the current ingress-nginx footprint	Envoy Gateway, APISIX, and other richer options remain eligible
T2	Initial sizing may use estimates; benchmarking is deferred	Estimates are planning values, not acceptance evidence
T3	Advanced generic limits should be provided by the gateway where practical	Application-specific GraphQL authorization and cost controls still remain in the application
T4	Equivalent dashboards may replace the existing dashboards	Metrics, logs, and traces matter more than dashboard continuity

Migration Decisions

ID	Decision	Status
D1	Put the new gateway on the original MetallB IP	✓ Agreed direction
D2	Forward unmatched traffic from the new gateway to ingress-nginx	✓ Agreed migration mechanism
D3	Migrate routes incrementally, with Keycloak after lower-risk routes	✓ Agreed sequencing
D4	Keep ingress-nginx only until fallback traffic reaches zero	✓ Retirement criterion
D5	Evaluate and document first; do not change the clusters yet	... Current phase

! External driver: ingress-nginx was retired in March 2026 and no longer receives fixes.

Current Shape

Two representative real clusters were inspected without recording private hostnames or addresses:

Item	Cluster A	Cluster B
Ingress resources	458	378
Rewrite annotations	447	367
CORS-enabled routes	218	178
ingress-nginx replicas	3	3
Observed CPU total	27m	42m
Observed memory total	517 MiB	635 MiB
Reserved CPU total	300m	300m
Reserved memory total	2304 MiB	2304 MiB

Current request path:

```

Wildcard DNS
  |
  v
MetalLB fixed IP
  |
  v
ingress-nginx
  |
  +--> TLS and redirects
  +--> regex rewrites and forwarded prefixes
  +--> CORS
  +--> oauth2-proxy and Basic Auth
  +--> application Services

```

Gateway API Model

HTTPRoute is a routing API, not a proxy. A Gateway API controller is still required.

```

GatewayClass --> selects the controller
  |
Gateway      --> owns listeners, addresses, and TLS
  |
HTTPRoute    --> matches host/path/header and selects backends
  |
Policy/CRD   --> authentication, limits, rewrites, and observability

```

Prefer standard Gateway API resources. Use implementation-specific policies only where the standard does not cover existing behavior.

Candidate Summary

Legend: ● preferred · ● retained alternative · ● evaluated · ● out of scope.

Status	Candidate	License	Gateway API	✓ Main benefit	! Main concern
●	Envoy Gateway	Apache-2.0	Strong conformance	Broad OSS security and observability policies	Higher resource use
●	Traefik Proxy	MIT	Core plus selected extensions	Small, simple deployment	More vendor CRDs; advanced shared limits need extra components
●	Apache APISIX	Apache-2.0	Partial support	OSS GraphQL-aware plugins	More components and lower Gateway API maturity
●	NGINX Gateway Fabric	Apache-2.0	Strong conformance	Familiar NGINX data plane	JWT and OIDC require NGINX Plus
●	HAProxy Ingress	Apache-2.0	Conformant implementation	Efficient data plane	Smaller policy and Gateway API ecosystem
●	Cilium Gateway API	Apache-2.0	Strong conformance	Integrated eBPF networking	Requires replacing Flannel; out of current scope

Resource Estimates

These are planning estimates, not benchmark results. They assume three data plane replicas, about 800 routes, metrics, logs, and tracing.

Candidate	Suggested reserved CPU	Suggested reserved memory
Traefik	300m	768 MiB
Envoy Gateway, local policies	500-700m	2.5-3 GiB
Envoy Gateway, global rate limiting	800m-1500m	3.5-4.5 GiB
NGINX Gateway Fabric	500-800m	2-3 GiB
APISIX with production dependencies	1-2 CPU	4-6 GiB

Proposed Envoy baseline:

Component	Replicas	Per replica	Approximate total
Envoy proxy	3	100m / 512 MiB	300m / 1536 MiB
Shutdown manager	3	10m / 32 MiB	30m / 96 MiB
Envoy Gateway controller	2	100m / 512 MiB	200m / 1 GiB
Total			530m / 2656 MiB

Repository policy requires memory requests and limits to be equal when these workloads are implemented.

GraphQL Constraints

- HTTP queries and mutations can be limited by identity, address, route, request count, body size, concurrency, and duration.
- Subscription clients use `graphql-transport-ws`.
- The subscription JWT is sent in the WebSocket `connection_init` message, after the HTTP upgrade. The gateway cannot authorize it from the handshake alone.
- JWT validation and role authorization must remain in the GraphQL application.
- Query depth, list cardinality, resolver fan-out, and true execution cost must be enforced in the application.
- APISIX `graphql-limit-count` charges HTTP POST requests by AST depth, but it does not cover subscriptions and depth is not a complete cost model.

Proposed Architecture

Use Envoy Gateway as the primary candidate. Keep Traefik as the low-resource fallback and APISIX as a future GraphQL-specialist candidate.

Target state:

Wildcard DNS



MetaLB fixed IP



Envoy Gateway (3 proxies)



Application Services

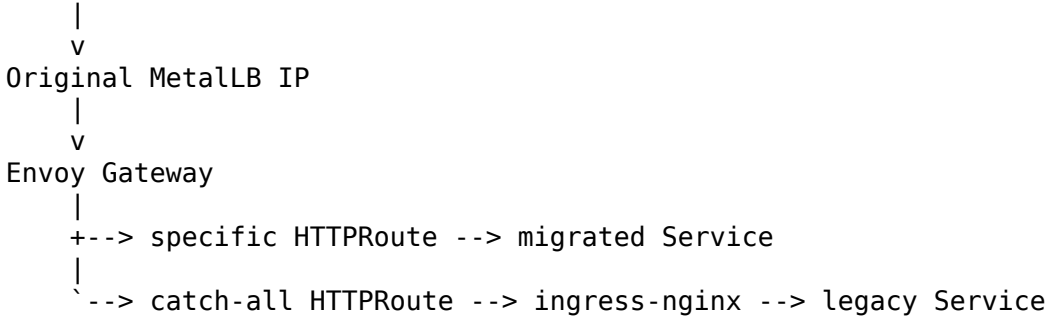
Initial policy placement:

Concern	Placement
TLS termination and redirects	Gateway/listeners and HTTPRoute
Host/path routing	HTTPRoute
Regex rewrites	Envoy Gateway extension policy
Basic Auth and external auth	Envoy security policy
JWT pre-validation	Envoy where possible; application remains authoritative
GraphQL roles and subscription authentication	Application
GraphQL complexity	Application
Initial rate limiting	Local Envoy policy
Exact shared quotas	Global rate-limit service and Redis, if justified

Migration Without DNS Changes

The new gateway takes the existing fixed IP and forwards all unmigrated traffic to ingress-nginx.

Wildcard DNS



Specific hostname routes take precedence over the hostname-less catch-all. The catch-all references the ingress-nginx Service through a ReferenceGrant.

Migration Sequence

1. Deploy Gateway API and Envoy Gateway on a secondary MetalLB IP.
2. Configure TLS and the catch-all route to ingress-nginx.
3. Test with client-side hostname resolution against the secondary IP.
4. Verify redirects, client IPs, WebSockets, large bodies, auth, and tracing.
5. Move ingress-nginx to the secondary IP.
6. Assign the original IP to Envoy Gateway.
7. Add specific HTTPRoutes incrementally; unmatched traffic keeps using NGINX.
8. Migrate Keycloak only after proxy headers, cookies, redirects, and token flows are verified.
9. Migrate each oauth2-proxy route together with its protected application.
10. Remove the catch-all and ingress-nginx after fallback traffic reaches zero.

The MetalLB IP handover is not atomic. Plan a short maintenance window and a tested reverse handover.

Fallback Safety Requirements

TLS

Envoy terminates TLS for migrated and fallback traffic. Before IP handover:

- inventory certificate and SAN coverage;
- confirm wildcard coverage for every hostname level;
- confirm listener certificate limits and cross-namespace references;
- preserve the existing external hostname and issuer for Keycloak.

Forwarded Headers

The fallback hop is HTTP unless backend TLS is explicitly configured. ingress-nginx must trust forwarded headers only from the Envoy proxy network:

```
use-forwarded-headers: true
proxy-real-ip-cidr: <exact trusted proxy CIDR>
```

Do not use 0.0.0.0/0. Incorrect trust can spoof client identity; missing trust can create HTTPS redirect loops.

WebSockets

- Preserve Host, Upgrade, Connection, and graphql-transport-ws.
- Configure long or disabled idle timeouts for subscriptions.
- Keep both paths available while old connections drain.
- Do not retry GraphQL mutations automatically.

Observability

Fallback requests cross two proxies and may be counted twice.

Client --> Envoy span --> NGINX span --> Application span

- propagate traceparent and request IDs;
- identify direct and fallback traffic in metrics and logs;
- apply rate limiting and retries at only one proxy layer;
- use fallback request volume as the retirement signal for NGINX.

Decision Gates

Implementation should not begin until all of these pass:

- ☐ Gateway API and controller version compatibility is pinned.
- ☐ TLS coverage is complete.
- ☐ The fallback preserves host, scheme, client IP, and WebSockets.
- ☐ No redirect loop occurs on TLS-enabled legacy Ingress resources.
- ☐ Large GraphQL and S3 requests pass configured limits.
- ☐ Keycloak issuer, callbacks, cookies, and token refresh remain unchanged.
- ☐ Three proxy replicas fit the selected resource budget under route load.
- ☐ The MetalLB IP handover and rollback are rehearsed.

References

- [ingress-nginx retirement](#)
- [Gateway API implementations](#)
- [Gateway API HTTPRoute](#)
- [Envoy Gateway security](#)
- [Envoy Gateway observability](#)
- [Envoy Gateway regex rewrites](#)
- [Traefik Gateway API](#)
- [APISIX Gateway API support](#)
- [APISIX GraphQL depth limiting](#)
- [ingress-nginx forwarded headers](#)